SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA0509**

COURSE OUTCOME COVERED

CO1: *Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 20%

QUESTIONS		
Q.No	**Question**	**Bloom's Level**
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

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Code of Conduct:

*I **SANDRA SUDEVAN(192424422)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

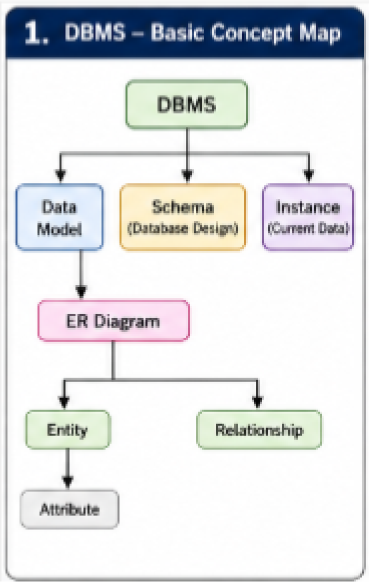
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. DBMS – Basic Concept Map

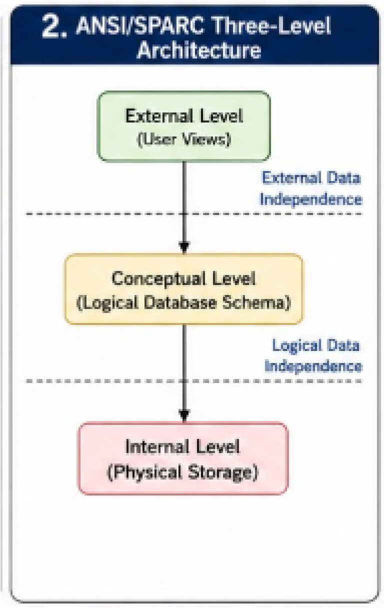


Explanation:

- **DBMS (Database Management System)** is software used to create, manage, and retrieve data efficiently.
- A **Data Model** defines how data is structured and organized.
- A **Schema** is the logical design or blueprint of the database.
- An **Instance** is the actual data stored in the database at a particular time.
- An **ER Diagram** is used to design the database structure.
- An **Entity** represents a real-world object (e.g., Student, Employee).
- An **Attribute** describes an entity (e.g., Name, Age).
- A **Relationship** shows how two or more entities are connected.

Example:
Student (**Entity**) → Roll No (**Attribute**) → Enrolls (**Relationship**) → Course (**Entity**)

2. ANSI/SPARC Three-Level Architecture



Explanation:
The ANSI/SPARC architecture separates the database into three levels:

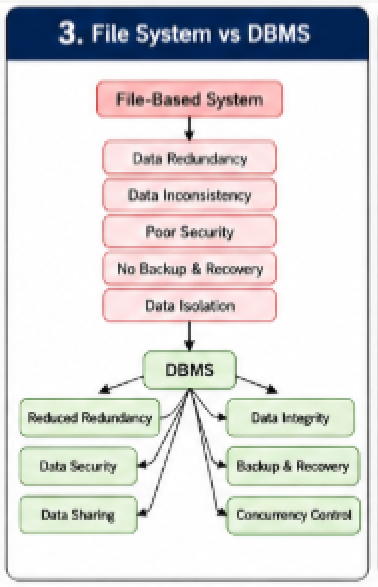
- **External Level:** User-specific views of the database.
- **Conceptual Level:** Overall logical structure of the database.
- **Internal Level:** Physical storage of data on the disk.

Data Independence

- **Logical Data Independence:** Changes in the conceptual schema do not affect user views.
- **Physical Data Independence:** Changes in physical storage do not affect the logical schema.

Benefit: Easier maintenance and flexibility.

3. File System vs DBMS



Explanation:

File System

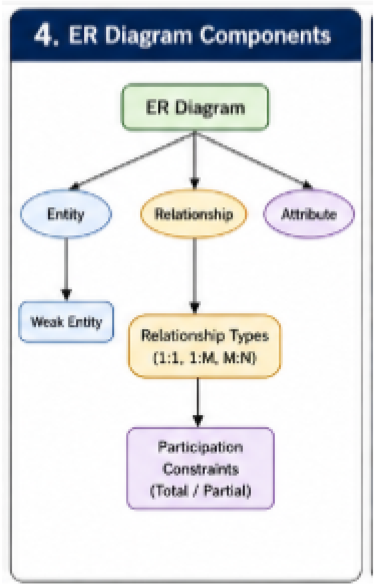
- Stores data in separate files.
- High data redundancy.
- Data inconsistency.
- Difficult to share data.
- Poor security.
- No proper backup and recovery.

DBMS

- Stores data in a centralized database.
- Reduces redundancy.
- Maintains consistency.
- Provides security.
- Supports backup and recovery.
- Allows concurrent access by multiple users.

Conclusion: DBMS is more efficient and reliable than the traditional file system.

4. ER Diagram Components



Explanation:

An ER Diagram is used for database design.

Components

- **Entity:** Real-world object.
- **Weak Entity:** Depends on another entity for its existence.
- **Attribute:** Property of an entity.
- **Relationship:** Association between entities.

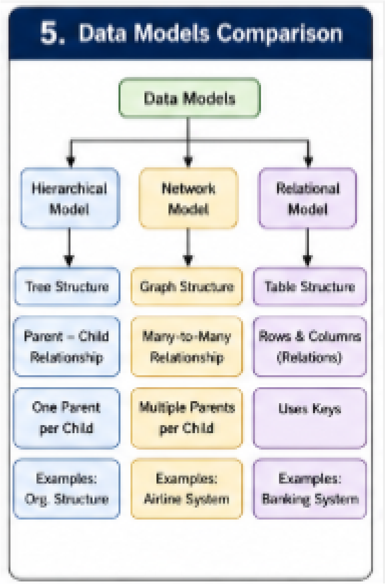
Relationship Types

- **1:1** – One employee manages one department.
- **1:M** – One department has many employees.
- **M:N** – Many students enroll in many courses.

Participation Constraints

- **Total Participation:** Every entity must participate.
- **Partial Participation:** Participation is optional.

5. Data Models Comparison



Explanation:

Hierarchical Model

- Organizes data in a tree structure.
- One parent can have many children.
- Simple but inflexible.

Example: Organization hierarchy.

Network Model

- Organizes data as a graph.
- One child can have multiple parents.
- Supports many-to-many relationships.

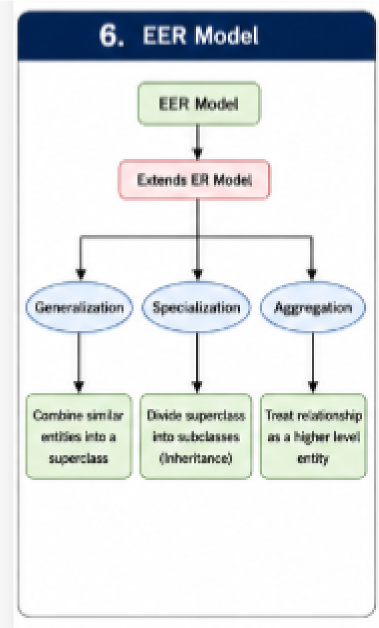
Example: Airline reservation systems.

Relational Model

- Stores data in tables (rows and columns).
- Uses SQL.
- Most widely used model today.

Example: Banking and college databases.

6. EER Model



Explanation:

The **Enhanced Entity-Relationship (EER)** model extends the ER model by adding advanced concepts.

Generalization

- Combines similar lower-level entities into one higher-level entity.

Specialization

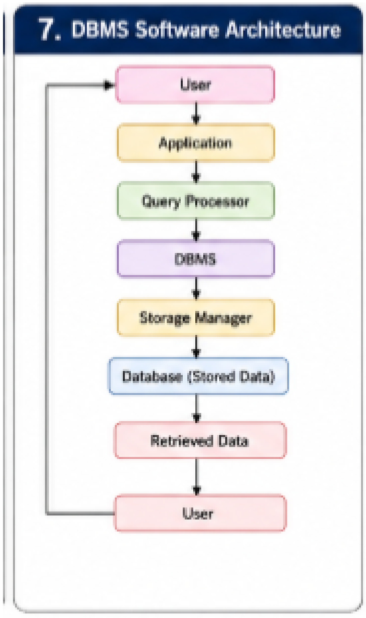
- Divides a general entity into specialized sub-entities.

Aggregation

- Treats a relationship as a higher-level entity.

Example:
Vehicle → Car, Bike (Specialization)

7. DBMS Software Architecture



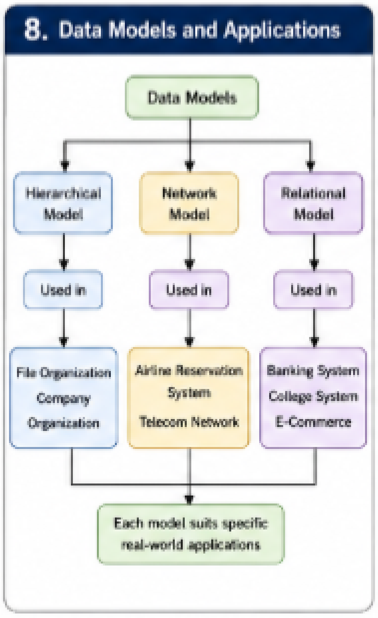
Explanation:

Flow of data in a DBMS:

1. User sends a request.
2. Application forwards the request.
3. Query Processor interprets SQL queries.
4. DBMS manages database operations.
5. Storage Manager accesses stored data.
6. Database returns the requested data.
7. User receives the result.

Purpose: Ensures efficient processing and retrieval of data.

8. Data Models and Applications



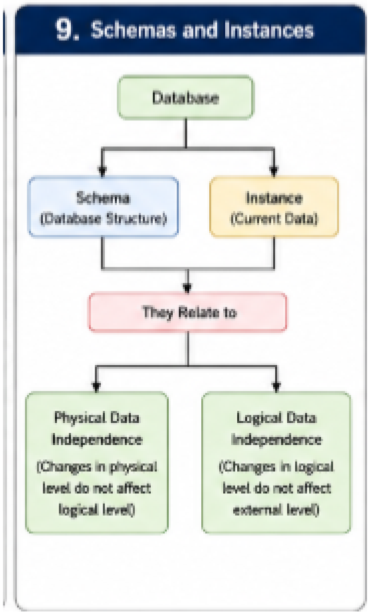
Explanation:

Different data models are suitable for different applications.

Data Model	Application
Hierarchical	Organization charts, File systems
Network	Airline reservation, Telecom networks
Relational	Banking, College management, E-commerce

Conclusion: The choice of data model depends on the application's requirements.

9. Schemas and Instances



Explanation:

Schema

- Database blueprint.
- Rarely changes.
- Defines tables, relationships, and constraints.

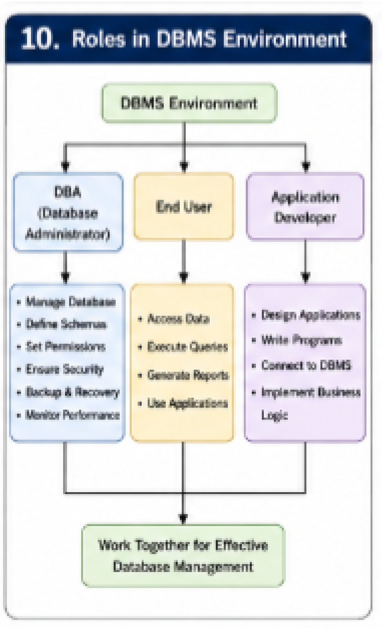
Instance

- Actual data stored at a given moment.
- Changes frequently.

Relation to Data Independence

- **Physical Data Independence:** Storage changes do not affect schema.
- **Logical Data Independence:** Logical changes do not affect user applications.

10. Roles in DBMS Environment



Explanation:

Database Administrator (DBA)

- Creates and maintains databases.
- Manages security and permissions.
- Performs backup and recovery.
- Monitors database performance.

End User

- Uses the database.
- Executes queries.
- Generates reports.
- Retrieves information.

Application Developer

- Develops database applications.
- Writes programs.
- Connects applications to the database.
- Implements business logic.

Conclusion: All three roles work together to ensure efficient and secure database management.